

Technical Handoff — Addendum

Fixes, an API-detection rewrite, and corrections to the 2026-08-09 handoff

Field	Value
Supersedes	Build 86fc41b8, dated 2026-08-09
This build	08f0109f, dated 2026-08-10
IPC surface	101 handlers, 101 bridged (was 99/99)
Automated checks	715 assertions across 9 suites
Scope	3 reported faults, 1 subsystem rewrite, 7 audit findings

This addendum records what changed and, where the original handoff was wrong, says so plainly. The original remains accurate on everything not listed here. Section 27 of the original is a ledger of known gaps; this document is its continuation.

1. Corrections to the original handoff

Three statements in the 2026-08-09 document did not match the code.

Where	What it said	What is true
Appendix B	"The database is a hint, never an override. What the binary actually imports wins."	The reverse. <code>inspectGame()</code> lets a database match replace the detected API outright. The behaviour is right — a remaster can change renderer without changing imports — but the description was backwards.
\$10	"Nothing is mirrored, repackaged or bundled, with one documented exception" (Legacy Geo3D).	There are two bundled trees. <code>bundled/vrexport/</code> is a live <code>bundledFallback</code> for <code>SuperVrExport</code> and <code>GeoVrExport</code> , and ships <code>geod3d9.dll</code> .
\$25, \$28	21 suites, 71,000+ assertions, <code>harness/validate.sh</code> as the release gate.	No harness shipped in the handed-over archive. A new one is included: 6 suites, 646 assertions, covering the changed subsystems and reconstructing the intent of the mod/layout, conflict-ordering and config matrices. It is not a like-for-like rebuild of the 71,000-assertion combination space.

2. Fault: VR-export links pointed at the wrong repositories

The Mods page linked `SuperVrExport` at `BlueSkyDefender/SuperVrExport` and `GeoVrExport` at `Flugan/Geo3D-Installer`. Neither is the download source. Both add-ons ship from one repository, which `CORE_SOURCES` has always named correctly.

<https://github.com/BerZerker96/Super-VRExport-Addon>

This is the same failure the original records for the Osiris VR Viewer link: a wrong URL in the Mods tab is invisible until somebody clicks it. The smoke suite now asserts the Mods page and the download source agree, in both directions.

3. Fault: "Remove all mods" did nothing

Two independent faults, either sufficient alone.

a. A call to a function that does not exist

`openDetail()` was invoked in two places in the renderer and defined in none. The call was the final statement of `uninstallAllMods()`, so the function threw a `ReferenceError` after running the removals but before refreshing the library or reporting anything. Nothing on screen changed.

b. The wrong identifiers

The loop iterated `g.inst`, which holds the UI's card ids (`supervr`, `geovr`), and passed them to a backend that keys the manifest by registry ids (`supervrexport`, `geovrexport`). Those calls matched no record — and the old `uninstall()` answered “nothing recorded” with `{ ok: true }`, so the loop counted them as removed while deleting nothing.

The fix. The operation moved into the main process as `installer.uninstallAll(game)`, over a new IPC channel. It reads the manifest — the only thing that knows what was placed — orders guests before hosts, removes each mod independently so one failure cannot abort the rest, and returns a real report. The renderer re-derives the game's state from disk afterwards.

4. Fault: per-mod “Uninstall” did nothing

`uninstall()` returned `{ ok: true }` when the manifest had no record for the id. The UI took that as success and dropped the card, but nothing was deleted — so the next folder rescan detected the mod still on disk and put it straight back. Three outcomes are now distinguished:

Outcome	Meaning	Behaviour
installed	Files were recorded	Deleted; record removed
adopted	Registered but installed by hand	Record removed; nothing deleted
untracked	Nothing recorded for this id	<code>ok: false</code> , <code>untracked: true</code> , with an explanation

It also accepts the UI's card ids defensively, deletes deepest paths first and removes directories it emptied (only when genuinely empty), reports per-file failures with a translated `errno`, and takes locked add-ons off with their host while leaving shared hosts in place.

Re-adoption guard. `refreshDetected()` auto-adopts anything found on disk. Called straight after a removal it re-adopted leftover signatures and the mod reappeared. Every uninstall path now passes `{ adopt: false }`.

5. Per-game tracking

The manifest at `<game>\.stereoscope\manifest.json` was always the record of what the app placed. What was missing was a way for the UI to read it rather than keep a parallel guess. `installer.installedMods(game)` is new and exposed over IPC; “Remove all mods” now builds its confirmation list from it, so what the user is shown is exactly what will be removed — including which entries are adopted and will only be de-registered.

6. Render-API detection, rewritten

Games were frequently reported with the wrong API. The detector read 24 MB of the executable and asked whether the bytes contained “`d3d11.dll`”. A byte sequence is not a dependency: the name appears in error strings, embedded resources, statically linked libraries and crash-reporter tables.

Detection is now modelled on how ReShade decides. ReShade hooks the loader and watches which graphics entry point the process really calls. The static equivalent of “what does this binary actually call” is the PE import directory, which the new `src/peimports.js` parses — both the classic table and the delay-load table.

Two failure modes this removes

Symptom	Cause
Every DX12 game also claimed DX11, and often led with it	<code>dxgi.dll</code> was counted as proof of DX11. DX10, DX11 and DX12 all import DXGI, as do many Vulkan titles.
DX12 titles showed no DX12 evidence at all	Modern engines delay-load <code>d3d12.dll</code> . The delay-load directory was never read as a structure.

Evidence model

Score	Evidence
4	An imported function unique to one API — <code>D3D12CreateDevice</code> , <code>vkCreateInstance</code> , <code>Direct3DCreate9</code>
3	A direct import of the API's own DLL
3	A delay-load import of the API's own DLL — the loader honours both, so it weighs the same
2	The engine runtime beside the exe imports it (a Unity exe imports no API itself; <code>UnityPlayer.dll</code> does)
2	The D3D12 Agility SDK shipped with the game (<code>D3D12Core.dll</code>)
1	A real sibling DLL in the game folder
1	The executable's own name — <code>Control_DX12.exe</code>

`dxgi.dll` now contributes no score. It is recorded only as a flag meaning “a DX10/11/12-class renderer is present”, and resolves to DX11 only when nothing stronger is found anywhere. It can never outrank real DX12 or Vulkan evidence, and never drags a DX9 game upward.

Files this app installed, and any known wrapper name, remain excluded: geo-11 places `d3d11.dll` beside the exe, and counting it would flip a DX9 game to DX11 on the next rescan. The old string scan survives only for binaries with no readable import table (packed or protected) and is scored below every real signal. Full rationale: [docs/API-Detection.md](#).

7. Defaults were overwriting user settings

Found by the new configuration suite. §17 of the original guarantees that installing “seeds missing sections so settings are editable before first launch, but never replaces a value the user has already set”. The code did not do that: `_installInner` wrote the whole DEFAULTS block unconditionally, so a hand-tuned configuration was replaced with the app's defaults on first install. Defaults now seed only the keys that are absent, and the pre-install snapshot of the user's values is taken on every install rather than only on an update.

8. Audit findings fixed alongside

Area	Finding
<code>src/scanner.js</code>	pickMainExe was defined twice. The second definition silently won and had lost the <code>JUNK_DIR</code> guard, so the fallback descended into <code>_Redist</code> , <code>installer_files</code> and <code>tools</code> — the folders §7 exists to keep out of the search. Duplicate removed.
<code>src/installer.js</code>	<code>apiOverrideNote</code> residue from a blanket string replacement: a <code>typeof</code> -guarded spread pasted into six places, two of them in <code>uninstall()</code> where the variable is not in scope and the guard always evaluated to an empty list. Removed.
<code>renderer</code>	The empty-scan guard in <code>realScan()</code> was duplicated; the second copy was unreachable. Removed.
<code>renderer</code>	Settings still told users their settings are “saved next to the app when that folder is writable (portable)”. They moved to the per-user data folder precisely because an update replaced that folder and destroyed the library. Corrected.

Area	Finding
src/store.js	Its file header described the same removed portable behaviour. Corrected.
src/mods.js	The geo-11 catalogue note read “Official HelixMod build v0.7.9” while the pinned url and version are v0.7.10. Corrected.

9. Test harness

The archive shipped without the harness the original describes, so nothing was runnable. A new one covers the changed subsystems and reconstructs the intent of the original combination matrices — 646 assertions, all passing.

```

bash harness/validate.sh      # all suites
node harness/smoke.js        # structure, IPC symmetry, registry, links, undefined globals
node harness/apidetector.js  # API detection against real synthetic PE images
node harness/uninstall.js    # install/uninstall lifecycle and per-game tracking
node harness/matrix.js       # every mod x every engine layout, and every output format
node harness/conflicts.js    # proxy slots, install order, ownership, uninstall permutations
node harness/configs.js      # config round-trips, unknown keys, seed-without-overwrite

```

Suite	Checks	Covers
smoke	94	Parsing, 101/101 IPC symmetry, registry integrity, catalogue links, bundled payloads, game database, build stamp, and that no renderer function is called that is never defined
apidetect	22	The PE import parser, then every detection regression against real synthetic PE32/PE32+ images
uninstall	41	Install → record → remove → verify, locked add-ons, adopted de-registration, untracked honesty, remove-all
matrix	346	88 mod/layout combinations and all 47 mod/output combinations. Every install reaches a decision, and every refusal AND acceptance is justified against MOD_API. Placement must land beside the real exe, not the game root.
conflicts	110	8 mod pairs in BOTH install orders, proxy-slot ownership, ReShade rehoming to dxgi, uninstall permutations with survivors verified complete after each removal, and no drift across repeated cycles.
configs	33	Round-trips, comment and unknown-key preservation, tuning surviving a mod upgrade, and defaults seeding without overwriting.

harness/pebuild.js writes genuinely valid PE32 and PE32+ images with a chosen import and delay-import table. This matters: detection reads the real import directory, so fixtures built from strings would test precisely the thing that used to be wrong. harness/uninstall.js drives the real installer against a sandboxed game folder using manual-core, so it needs no network.

The undefined-global check was verified by deleting the openDetail definition and confirming the suite fails.

10. Re-audit

Every fix above was re-verified from the packaged archive rather than the working copy, with three suites written to attack the fixes rather than confirm them. wiring.js asserts the whole chain — markup onclick, integration-layer definition, IPC channel, preload binding, ipcMain handler — because both original faults were wiring faults, not logic faults. uninstall-e2e.js drives uninstall through the renderer’s exact serializeGame payload and bidOf id mapping rather than calling the backend directly, plus the edge cases: files deleted by hand, adopted and installed mods removed together, nested executable layouts, an undeletable file, double uninstall, and a game with no folder. apicompare.js reimplements the pre-fix detector verbatim and runs both head to head.

API detection, measured

Detector	Correct	Rate
Old (string scan)	8 / 14	57%
New (PE import directory)	14 / 14	100%

Six cases fixed, zero regressions. The failures removed: a DX9 title mentioning d3d11.dll in an error string (was DX11); an OpenGL title with a dxgi reference in a bundled library table (was DX11); a DX11 title with a d3d12 mention in a feature-check string (was DX12); a DirectDraw title mentioning d3d9 in a compat note (was DX9, not DX7); a Unity stub whose UnityPlayer.dll is DX12 (was DX11); and a DX9 title with geo-11 already installed beside it (was DX11). The common thread is that a byte match is not a dependency — every one is a binary that MENTIONS an API it does not use.

11. What is still open

Item	Status
The larger combination suites from §25 (exhaustive, total, everymod, fullmatrix, exematrix...)	Partially reconstructed. matrix/conflicts/configs cover the mod-by-layout, ordering and config spaces; the 103 real executable shapes (exematrix) and the 80-mod 6DOF sweep (everymod) are not rebuilt.
Hardcoded LATEST/OLDER version tables in the renderer	No longer displayed as fact. Version labels render through verLabel(), which shows a dash until the value has been confirmed against the network. The tables themselves remain as mockup seeds and should go when that layer is retired.
The update .bat swap, RAR extraction, elevated 3DVision4All helper	Still verified by inspection only, as recorded in §27 of the original. Unchanged by this work.
Windows runtime verification of these fixes	The suites run on Linux. The uninstall paths are exercised end to end against a real filesystem, but a Windows pass on a real library is the remaining confirmation.

End of addendum. Build 08f0109f. Generated 2026-08-10.